8. Bully algorithm for distributed database.

→ a) Need & Importance of the algorithm

In a distributed database system, data is spread across multiple nodes (sites), and certain operations require coordination such as transaction management, concurrency control, replication control, and recovery.

For these tasks, one node usually acts as a coordinator (leader)

Why a coordinator is needed?

- To manage distributed transactions
- To resolve conflicts (lock conflicts)
- To synchronize replicated databases.
- To handle failure recovery.

The problem:

- Node can fail unexpectedly
- The coordinator itself may crash.
- The system must continue functioning without manual intervention.

Importance:

- Elect a new coordinator automatically
- Ensure fault tolerance.
- Maintain system consistency.
- Minimize downtime in distributed db's.

b) Description of the Bully Algorithm:

The Bully Algthm is a leader election algthm. used in distributed systems where:

- Each process has a unique ID.
- Higher ID ⇒ higher priority.
- All nodes know the IDs of other nodes.

The process with the highest ID among active processes becomes the coordinator.

Algorithm steps:

1. Failure detection:
   A process detects that the coordinator is not responding.
2. Election initiation:
   The detecting process sends an ELECTION message to all processes with higher ID's.
3. Response handling:
   - If no higher-ID process responds → The initiating process becomes the new coordinator.
   - If any higher-ID process responds → The initiating process stops & waits.
4. Recursive Election:
   The higher-ID process now starts its own election
5. Coordinator announcement:
   The final winner broadcasts a COORDINATOR message to all processes.

c) Real-time application of Bully algorithm:

1. Distributed management systems:
   - selecting a transaction coordinator
   - Managing replica synchronization.
   - Handling distributed locking.
2. Cloud computing & Microservices.
   - Leader election in service orchestration
   - Failover management in distributed services.

3. Fault-Tolerant Systems:
- Automatic recovery after node failure.
- Ensuring uninterrupted service availability.

4. Networked Systems:
- control ~~node~~ election in distributed monitoring systems.

d) Requirements to implement the Bully algorithm in a distributed database system:

*System Requirements:

1. Unique process IDs each database node must have a unique identifier
2. Reliable communication channel, message passing must be possible between nodes.
3. Failure detection mechanism.
   Time-out based or heart beat-based failure detection.
4. synchronous or semi-synchronous network
   Message delays should be bounded.

*Software Requirements:

- Message types:
  - ELECTION
  - OK
  - COORDINATOR.
- Timeout handling logics
- Leader state management.

* Assumptions:

- Nodes can crash but do not behave maliciously
- Failed nodes may later recover.
- Network partitions are not permanent.